Set:B

BRAC UNIVERSITY

Department of Computer Science and Engineering

Examination: Mid Term
Duration: 1 Hour 20 Minutes

Semester: Summer 2025
Full Marks: 25

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin is part of the examination.

[redacted]

1. CO1 A processor executes a program, which has a total of 30×10^9 instructions, categorized as follows:

Instruction Type	Instruction Count	Cycle per instruction
R-type	50% of total instructions	2
I-type	20% of total instructions	4
SB-type	20% of total instructions	3
U-type	10% of total instructions	1

a) Find the average CPI of the program. 2

b) If the CPU clock frequency is 3.2 GHz, calculate the execution time of the program. 1

2. CO1

a) Define throughput and response time in computer systems. What does it mean for a system to have high throughput with high response time? 2

b) A program runs on CPU M in 100s. CPU N optimizes 60% of the instructions to run 1.5 times faster, while the remaining instructions execute at the same speed. Calculate the new execution time of CPU N. 2

3. CO1 Read each statement carefully. Determine if the following sentences/statements are true or false. For any false sentence/statement, write its correct form.

a) - 0xB,4A3 can be stored precisely in the following format 1

Sign (1 bit)	Exponent (5 bits)	Fraction (10 bits)

b) If a CPU with 2.5x instruction count and 4GHz clock speed performs the same task in 10 seconds as another CPU with 1.5x instruction count and 4.5GHz clock speed, then the CPI of the first CPU must be lower. 1

c) A 16-core processor can always achieve a 16× speedup over a single-core version on any program. 1

4. CO1 $(1.1111011011)_2 \times 2^{17}$; You want to store this floating point number using IEEE754 floating point representation.
You have two format options:

	Sign Bit	Exponent	Fraction
Option - 1	1 bit	7 bits	6 bits
Option - 2	1 bit	6 bits	7 bits

a) *If your goal is to achieve precision, which of the two options would you choose? Justify your answer.* 2

b) *What is the range of usable actual exponent that can be stored in a 24-bit IEEE format where 6 bits are reserved for the fraction?* 2

6 bits

5. CO2 a) Construct the equivalent RISC-V assembly code for the following C program: 5

```
arr[18] = func(8, 16);
c = arr[19] + d

int func(int a, int b) {
    if (a <= b) {
        while (a != b) {
            p += 2;
        }
        b = b & 1;
    } else {
        p = b << 4;
    }
    return (p + b);
}
```

Variables c and d are stored in registers X_{18} and X_{19} respectively.

Function arguments a and b are stored in registers X_{10} and X_{11} respectively.

Variable p in the function func, also uses register X_{19}.

arr is a halfword array. The base address of arr is stored in X_{20}.

Hint:
<< means left shift

slli

b) 3

Instruction Address	Instruction
0xB0A1834A	bne X_{21},X_{22} L0.
.....	.....
.....	.....
.....	.....
.....	.....
0xB0A1838A	L0: add X_1, X_4, X_7

$X_{22}, X_{22}, L0$.

$0x\,B0$

Convert the instruction, bne X_{21} ,X_{22} L0; to machine code. For bne, opcode = 1100111, funct3 = 001; Your final answer must be in Hexadecimal.

c) Construct the equivalent RISC-V assembly code for the following C program: 3

```
a = 0xD035B1BC
b = a / 32
```

a and b variables are stored in X_{19}, X_{20} registers, respectively.

a/32. 32 2^5.

slli

Question - 7:

You are analyzing a binary dump. At memory address `0x00004010`, you find the following machine code for a `blt` (Branch if Less Than) instruction: **`0xFEA2CE63`**.

Address Instruction

0x3FFC: addi x10, x10, 1

0x4000: sub x5, x20, x21

0x4004: slli x5, x5, 2

0x4008: add x6, x10, x5

0x400C: lw   x7, 0(x6)

0x4010: blt x5, x10, Target <-- Machine Code: 0xFEA2CE63

0x4014: sw   x7, 0(x6)

0x4018: addi x22, x22, 1